HALEEMA TALLAT

AI Engineer · Evaluations, agents, LLM inference
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AI SAFETY & EVALUATIONS

ConspiracyBench — Stationhouse, research collaborator · judge study 2026 – Ongoing

- Led the project's judge selection, a scalable-oversight question: built a 150-reply blind validity set with pre-registered scoring, coordinated 4 labellers, and compared 9 judges plus jury and cascade designs. The incumbent judge called divergence at nearly twice the human rate (43% vs 23%); the team's vote was based on these results.
- Found the headline divergence rate was inflated from about 20% to 49%: Qwen 3.5 4B spent its token budget on reasoning and returned empty answers on half its turns, which the judge still graded. Fixed the scorer to mark them invalid; another contributor reproduced the issue on two more judges.
- Fixed a parser bug that read negated labels ("does not diverge") as verdicts, and proposed the study's reliability bar.
- Built a log-analysis tool that shows where a run's time goes and checks byte for byte that speed changes leave prompts unchanged; it traced 51 timed-out requests per run to a concurrency default.

LeakBench — github.com/haleematallat/leakbench 2026

- Built a 30-task Inspect benchmark that tests whether AI agents can find and fix the bugs that make an ML evaluation report the wrong number, with 5 correct pipelines to catch over-editing and hidden checks in Docker sandboxes.
- Made every task verifiable and hard to reward-hack before any model ran: 120 no-model checks show each is solvable and fails if data is edited or tests are faked. Fixed ranges that rejected valid fixes or passed half-fixes.
- Showed it separates models: Qwen 3.5 9B solved 75% of bug tasks against 29% for the 4B (95% CI +29 to +61 points). Without a hint that the eval might be wrong, the 4B fell to 12%.

Technical AI Safety course — BlueDot Impact Sep 20 – Oct 24, 2026

ARENA curriculum (self-study) 2026

Talk: Sleeper Agents, Deceptive LLMs & Safety Training — GDG DevFest Lahore · video Dec 2024

EXPERIENCE

Senior AI Engineer, Agents & LLM Infrastructure — IgniteTech (via Crossover) Jul 2025 – Present

- Designed and built IRIS's first production social engine in ~3 months: 1,000+ customer interactions a day across 10–15 channels on multi-tenant Node.js/TypeScript services on AWS. It became the template for 20 more engines; scaled the platform to 27 on Kafka/KEDA and cut memory 6–10× and startup ~20× by replacing Java monoliths.
- Architected a distributed platform running up to 500 concurrent coding agents: LangGraph on Ray, Pydantic-typed state and tools, Kafka events, Redis and PostgreSQL state, on Kubernetes on AWS.
- Made agent runs measurable by tracing every model and tool call in Langfuse, tracking cost per run and tool-call errors in ClickHouse, and scoring drafted replies by acceptance and edit rates.
- Designed the governed MCP layer for Claude Code, Cursor and our agents (OAuth2, capability controls, human-in-the-loop approval for high-impact actions), and a rule-checking agent on top of it.

Computer Vision Engineer — Brick & Mortar AI Mar – Oct 2025

- Fine-tuned a ResNet-12 few-shot classifier on damaged barcode images, then showed it scored 89.7%, not the reported 46.7%, by fixing split leakage and protocol mismatches in the eval harness.
- Cut identity switches 20% with segmentation-masked ReID; GPU RTSP decoding raised throughput 10–15×.
- Took the tracking pipeline to production: PyTorch to ONNX to TensorRT, a custom Triton ReID backend with a re-tuned matching threshold, and fixes for shape-mismatch and runner-registration failures in the deployed stack.
- Built zone monitoring for retail clients to analyse customer behaviour and retention across store zones, and face recognition with facial landmarks for staff attendance and spotting returning customers.

Computer Vision Engineer — NextBridge Nov 2024 – Mar 2025

- Built a smart wheelchair's WebRTC control stack: signalling server (SDP/ICE), joystick sender, ROS 2 receiver.
- Made the link fail-safe: heartbeats, backoff reconnects, a stop-on-link-loss watchdog; measured round-trip latency.

Machine Learning Engineer — Powersoft19 Nov 2023 – Nov 2024

- Cut field engineers' problem-resolution time 90% with a hybrid RAG system over technical docs, tuned on recall@k.
- Cut fault-detection turnaround from ~48 hours to ~30 minutes by leading 3 engineers to replace manual inspection with anomaly detection on 1TB+ of device telemetry.
- Built an ML experimentation platform with CLI tools, monitoring dashboards and GitHub integration that let 200+ engineers deploy and track experiments, cutting setup from days to hours.

Research Engineer — KICS, Center for Language Engineering, UET Nov 2022 – Nov 2023

- Reached 98% accuracy with a distributed OCR pipeline for historical archives, running at 99.9% uptime.
- Reviewed the literature on OCR techniques for a multilingual LLM setup, and reproduced published RAG methods.
- Developed ticker-analysis code for Tajziyakar, an archive-scale OCR pipeline that flags TV news tickers.

EDUCATION & SKILLS

Education: BSc Electrical Engineering, UET Lahore (2023)
Awards: CureMD Master Inventor Award (2022) · COHACK by DevNation hackathon, 1st place (2022)
Evaluations: Inspect, benchmark and task design, LLM-as-a-judge validation, human–AI agreement, paired bootstrap
LLMs & agentic coding: Claude Code, Cursor, Codex, MCP, LangGraph, Anthropic API, LibreChat, RAG
Computer vision: YOLOv8/v11, segmentation, re-identification, face recognition, facial landmarks, zone monitoring
Model serving: vLLM, ONNX, TensorRT, Triton Inference Server, NVIDIA Jetson, CUDA
Languages & ML: Python, TypeScript, C, C++, SQL, Node.js, Next.js, React, PyTorch, JAX, fine-tuning